

# Zuodong, WANG

## 王作栋

Ningbo, China  
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wangzuodong1997.github.io

### RESEARCH INTERESTS

#### Numerical Analysis

Fluid dynamics, semi-linear parabolic problems, PDEs with stiff source term, structure-preserving schemes, a posteriori error analysis

#### Programming

C++, Python, Matlab, Julia

### ACADEMIC APPOINTMENTS

#### Postdoctoral Researcher (2025-present)

Eastern institute of technology, Ningbo [link](#)

- Project: Efficient high-order structure-preserving schemes for evolution problems
- Advisor: Jie Shen

### EDUCATION

#### PhD (2025)

CERMICS and SERENA team,  
Ecole nationale des Ponts et Chaussées, Institut Polytechnique de Paris & Inria, Paris

- Thesis title: Efficient numerical schemes for evolution equations with singularities and shocks
- Advisors: Zhaonan Dong, Alexandre Ern

#### Master (2022)

Ecole nationale des Ponts et Chaussées, ParisTech & Sorbonne Université

- Applied Mathematics: Modeling, Analysis and Simulation
- Grade: 16.10/20 (top class)

#### Engineer's degree (2021-2022)

Ecole nationale des Ponts et Chaussées, department IMI

#### Bachelor (2020)

Beijing University of Technology

- Mathematics and Applied Mathematics
- GPA: 3.77/4.0

### LANGUAGES

Chinese (Mandarin): Native,  
English: Fluent, French: Fluent, Japanese: Duolingo

### CONFERENCES

#### 11th International Conference on Computational Methods in Applied Mathematics

Vienna, Austria, 2026 – *invited talk*

#### Tsinghua Sanya International Mathematics Forum on Advanced numerical schemes for multiphysics problems

Sanya, China, 2025 – *invited talk*

#### 30th Biennial Conference on Numerical Analysis

Glasgow, UK, 2025 – *contributed talk*

#### International Conference on Applied Mathematics

Hongkong, China, 2024 – *contributed talk*

#### Finite Element Rodeo

Houston, USA, 2024 – *talk in mini-symposium*

#### The 6th SIAM Texas-Louisiana Sectional Meeting

Lafayette, USA, 2023 – *talk in mini-symposium*

### TEACHING

#### Numerical analysis (French)

2025 – *Paris Dauphine University*

#### Explorations in Mathematics (English)

2024 – *Texas A&M university*

### AWARDS

#### CSC Scholarship

Paris, 2020 – 2022

#### Outstanding Graduate

Beijing University of Technology, 2020

#### Model Student of Academic Records

Beijing University of Technology, 2016 – 2020

### HOBBIES

ACG(Anime, Comics, Games)

### EXPERIENCE

#### Visiting scholar

Texas A&M university

*College station, USA, September 2023 – May 2024*

- Project: Design, analyse and testing Invariant-domain-preserving schemes for hyperbolic problems

#### Internship (stage court)

Inria Centre de Paris

*Paris, France, June – September 2021*

- Project: Hybrid high-order method for elliptic PDEs on curved and complicated domains

## PUBLICATIONS

1. **Pressure-robust hp-a posteriori error estimates of H(div)-conforming discontinuous Galerkin methods for the Stokes equations**  
2026, Zhaonan Dong, [Zuodong Wang](#), Lina Zhao – *Submitted* [link](#)
2. **A high-order nodally bound-preserving and mass-conservative method for linear fourth-order elliptic problems and its applications to nonlinear parabolic equations**  
2026, Jie Shen, [Zuodong Wang](#) – *Submitted* [link](#)
3. **A posteriori error analysis and adaptivity of a space-time finite element method for the wave equation in second order formulation**  
2026, Zhaonan Dong, Emmanuil H Georgoulis, Lorenzo Mascotto, [Zuodong Wang](#) – *Accepted in Numerische Mathematik* [link](#)
4. **A priori and a posteriori error estimates of a  $C^0$ -in-time method for the wave equation in second order formulation**  
2026, Zhaonan Dong, Lorenzo Mascotto, [Zuodong Wang](#) – *Numerische Mathematik* [link](#)
5. **A bound-preserving scheme for the Allen-Cahn equation**  
2025, Zhaonan Dong, Alexandre Ern, [Zuodong Wang](#) – *Computers & Mathematics with Applications* [link](#)
6. **Mass conservative limiting and applications to the approximation of the steady-state radiation transport equations**  
2025, Jean-Luc Guermond, [Zuodong Wang](#) – *Journal of Computational Physics* [link](#)
7. **Asymptotic and invariant-domain preserving schemes for scalar conservation equations with stiff source terms and multiple equilibrium points**  
2024, Alexandre Ern, Jean-Luc Guermond, [Zuodong Wang](#) – *Journal of Scientific Computing* [link](#)
8. **Hybrid high-order methods for elliptic PDEs on curved and complicated domains**  
2022, Zhaonan, Dong, [Zuodong Wang](#) – *13th ICOSAHOM 2020/2021* [link](#)
9. **The Proof of 2n Circle Arrangement Conjecture**  
2017, [Zuodong Wang](#) – *Pure Mathematics* [link](#)